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Application for Emotion Recognition in Educational Context

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Abstract

In the present day, more and more researchers, businesses, and educators want to incorporate CNNs into their work, to increase productivity. This project aims to improve quality of communications in the educational context. While providing its users with an ability to on one hand – provide emotional feedback, and on the other hand – receive it, the application also allows to train a custom model, and upload one's own versions of the provided datasets.

On the market of emotion-tracking systems, there are no readily available analogues for this project. This is due to privacy concerns, as well as existence of the simpler solutions that do not involve CNNs (Convolutional Neural Networks). We address the privacy issues below, while the already existing solutions only pushed to create a better, and more accurate application.

The idea of the main part of the application is that during a class lecturers can get information about current feelings of their students via emotion recognition. The pictures of faces of students are analyzed and then deleted, not obscuring privacy. The emotion data is sent to the lecturer anonymously, thus the lecturer can only check the emotions, without knowing, what student he is inspecting.

Our project is not only of value for the industry, it is also usefull from the academic point of view. The thesis writing consisting mainly of three stages: getting access to the datasets, training models, while searching for the architecture that would yeild the best validation accuracy, and the third being the programming of the application that has user-friendly GUI, and is easy to install.

The datasets that we have been using for this research project are readily available online. This provides this paper a greater credibility, since other researchers are capable to derive same results as us. Same goes for the model-training scripts and application development frameworks. The libraries used for training and experimenting with emotion recognition were Pytorch and Tensorflow, which are both Python libraries, readily available on the Internet. The datasets used are of reasonable size. This allows to train faster, without utilizing excessive memory spaces. Not only the sizes are user-friendly, also the datasets themselves are available online. If one would want to take hold of a full version of a sizable dataset, such as AffectNet, and train a model on it, permissions may be required.

Streszczenie

W dzisiejszych czasach coraz więcej badaczy, przedsiębiorstw oraz instytucji edukacyjnych dykt do wykorzystania sieci neuronowych typu CNN w swojej pracy w celu zwiększenia produktywności. Celem niniejszego projektu jest poprawa jakości komunikacji w kontekście edukacyjnym. Aplikacja zapewnia użytkownikom możliwość, z jednej strony, przekazywania informacji zwrotnej dotyczącej emocji, a z drugiej strony ich odbierania. Dodatkowo umożliwia trenowanie własnego modelu oraz przesyłanie własnych wersji dostarczonych zbiorów danych.

Główna idea aplikacji polega na tym, że podczas zajęć dydaktycznych wykadowcy mogą uzyskiwać informacje o aktualnym stanie emocjonalnym studentów za pomocą rozpoznawania emocji. Zdjęcia twarzy studentów są analizowane, a następnie natychmiast usuwane, co pozwala na zachowanie prywatności. Dane dotyczące emocji są przesyłane do wykadowcy w sposób anonimowy, dzięki czemu może on analizować emocje bez możliwości identyfikacji konkretnego studenta.

Projekt ten ma wartość nie tylko z punktu widzenia przemysłu, lecz także z perspektywy akademickiej. Praca dyplomowa składa się zasadniczo z trzech etapów: uzyskania dostępu do zbiorów danych, trenowania modeli przy jednoczesnym poszukiwaniu architektury zapewniającej najwyższą dokładność walidacyjną oraz implementacji aplikacji posiadającej przyjazny dla użytkownika interfejs graficzny i łatwą w instalacji.

Zbiory danych wykorzystywane w niniejszym projekcie badawczym są publicznie dostępne w Internecie. Nadaje to pracy większą wiarygodność, ponieważ inni badacze są w stanie odtworzyć uzyskane wyniki. To samo dotyczy skryptów do trenowania modeli oraz frameworków użytych do tworzenia aplikacji. Bibliotekami wykorzystywanymi do trenowania i eksperymentowania z rozpoznawaniem emocji były PyTorch oraz TensorFlow – oba są bibliotekami języka Python, powszechnie dostępnymi w Internecie. Używane zbiory danych charakteryzują się rozsądnym rozmiarem, co umożliwia szybsze trenowanie modeli bez konieczności wykorzystywania nadmiernych zasobów pamięci. Co więcej, same zbiory danych są łatwo dostępne online. W przypadku chci wykorzystania pełnej wersji dużego zbioru danych, takiego jak AffectNet, oraz trenowania na nim modelu, może być wymagane uzyskanie odpowiednich zezwoleń.

Division of work

Theory work distribution	
Maksym Nosal	Section on FER-2013 dataset; section on ResNet as CNN architecture; statistics metrics (accuracy, recall, precision, F1-score, ROC); model training.
Oleksandr Shkil	Section on AffectNet dataset; section on ChatGPT as emotion recognition tool; RAM capacity and training time; experimental results chapter; Python libraries experiments; custom dataset compilation; model training.
Maciej Sosnowski	Section on RAF-DB dataset; sections on Visual Geometry Group and EfficientNetB0 architectures; LLM emotion recognition experiments; experiments with models; model training

Table 1: Division of work

Application work distribution	
Maksym Nosal	Work on the Educator side of the application; The custom model training part; the communication between Student and Educator; E2E tests.
Oleksandr Shkil	Application idea; Student side; Functional and Non-functional requirements; Tests of the application using the custom dataset
Maciej Sosnowski	Work on the Educator side; Overall enhancements; GUI; unit and integration tests; application architecture; OpenCV integration.

Table 2: Division of work on the application

Introduction

The primary challenge of this thesis research is to train Convolutional Neural Networks(CNNs) to be able to recognize students' emotions, to help the people teaching them to better their explanations. This task requires, however, that the end users get an application that is reliable, easy to use, and that brings a positive incentive into the learning process. This involves minimizing the effort, required to understand the application's User Interface, as well as making use of the data harnessed by the emotion recognition models.

There are other solutions, that help teachers and professors to better understand the sentiments of their apprentices. Nevertheless, for the most part, they are requiring students to manually provide the data of their emotional state. This process may be cumbersome for many students, and may lead to false conclusions. The applications that use CNNs are not yet widely spread, and mostly applied in research settings.

From our point of view, there are several interesting questions that our research touches. First of all, it uses publicly available tools. We use models and datasets, available for general public. This allows other researchers to recreate our findings. Another interesting thing for us is, indeed, how well CNNs perform, when tasked with recognizing emotions, including such of different regions, ethnicities, and ages, etc. The system should include components to collect data, analyze it, and produce useful statistics about the students' emotional states. The connections between different parts have to be highly optimized, to allow real-time data processing. The benefits of utilization of our product are that one will be able to improve the flow of education and permit for educators to produce more value.

To conclude this introduction, we would like to briefly describe, what is in each chapter. In the first chapter we discuss the main datasets we worked with, including their properties, methods of collection, etc. In the second chapter we discuss different Convolutional Neural Networks architectures. The next chapter speaks of metrics, by which we judge model's performance. The fourth chapter contains some of the application documentation. Finally, fifth chapter concludes the project, by recommending the best model for emotion Recognition.

Chapter 1

Datasets for Emotion Recognition

1.1 AffectNet

The AffectNet dataset is distinguished by its unique way of storing information about particular emotions, shown by images, see Figure 1.1. Namely, it uses both Categorical (only for 450'000 out of 1000'000) and Continuous valency-arousal labels. Categorical labels split the emotions between the 8 basic ones: happiness, sadness, anger, fear, disgust, surprise [2]. In the Continuous method of emotion notation, 2-dimensional space is used. In this case, each emotion is represented by a pair of values, one of which is representing arousal, which means that the event experienced by the person, showing the emotion is exciting, agitating or calming and soothing. The second axis, valence, depicts how positive or negative the event experienced is. Figure 1.1 shows an example of valence-arousal axis and emotions, mapped onto it.

The data for AffectNet was taken from the Internet, hence the name [1]. The procedure of data collection was as follows. The three search engines (Google, Bing, Yahoo) were queried in 6 different languages by the team of native speakers of these languages: English, Spanish, Portuguese, German, Arabic, Farsi [1]. The querying was done using 1250 emotion related tags. Additionally, negative tags such as "drawing", "cartoon", to maximally avoid non-human results. The results were then inspected by people, to assure sanity of the search results [1]. After return of 1,800,000 URLs were stored in the database. After that, OpenCV was used to find bounding boxes for each face. The table below shows the percentage of images that correspond to the queries that obtained them.

The Table 1.2 shows the numbers of each emotion gathered.

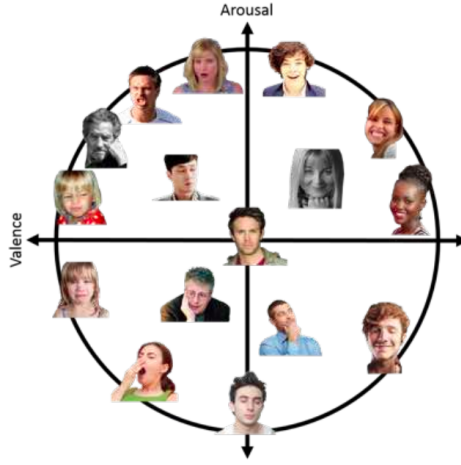


Figure 1.1: Sample images in Valence Arousal circumplex [1].

Table 1.1: Percentage of Annotated Categories for Queried Emotion Terms (%) [1]

Annotated Expression	Query Expression						
	HA	SA	SU	FE	DI	AN	CO
NE*	17.3	16.3	13.9	17.8	17.8	16.1	20.1
HA	48.9	27.2	30.4	28.6	33.0	29.5	30.1
SA	2.6	15.7	4.8	5.8	4.5	5.4	4.6
SU	2.7	3.1	16.0	4.4	3.6	3.4	4.1
FE	0.7	1.2	4.2	4.0	1.5	1.4	1.3
DI	0.6	0.7	0.7	0.9	2.7	1.1	1.0
AN	2.8	4.5	3.8	5.6	6.0	12.2	6.1
CO	1.3	0.9	0.4	1.1	1.1	1.2	2.4
NO	5.4	8.7	4.8	8.1	8.8	9.3	11.2
UN	1.3	3.1	4.3	3.1	4.1	3.7	2.7
NF	16.3	18.6	16.7	20.6	16.9	16.8	16.3

* NE, HA, SA, SU, FE, DI, AN, CO, NO, UN, and NF stand for Neutral, Happy, Sad, Surprise, Fear, Anger, Disgust, Contempt, None, Uncertain, and Non-face categories, respectively.

Table 1.2: Number of Annotated Images in Each Category [1]

Expression	Number
Neutral	80,276
Happy	146,198
Sad	29,487
Surprise	16,288
Fear	8,191
Disgust	5,264
Anger	28,130
Contempt	5,135
None	35,322
Uncertain	13,163
Non-Face	88,895

1.2 FER-2013

FER-2013 was introduced in a Kaggle challenge in parallel with the International Conference on Machine Learning. The Facial Expression Recognition (FER) dataset addressed a need for publicly available, large scale dataset to develop and benchmark facial expression recognition algorithms.



Figure 1.2: Sample images from the FER-2013 dataset, illustrating distinct emotion categories [4].

The data consists of 48x48 pixel greyscale images of faces. The faces have been automatically registered so that the face is more or less centred and occupies about the same amount of space in each image. In Figure 1.2 we can see some example data from the set. The training set consists of 28,709 examples and the public test set consists of 3,589 examples [3].

The task is to categorize each face based on the emotion shown in the facial expression into one of seven categories: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral) [3]. In the two figures: 1.3 and 1.4, one can see the distribution of images between emotions in the dataset.

This dataset is publicly available and focuses on a range of expressions in relatively uncontrolled conditions (the images were collected from the internet). Some of the uses of FER-2013 include: training and evaluating deep learning models, benchmarking new algorithms and exploring challenges in facial expression recognition.

In spite of popularity, the dataset has its limitations and challenges, specifically class imbalance, noisy labels, low image resolution and lack of context or metadata. Below is the explanation of before-mentioned problems [7].

The distribution of images across the seven emotion categories is not uniform. The 'happy' category, has a significantly larger number of samples compared to

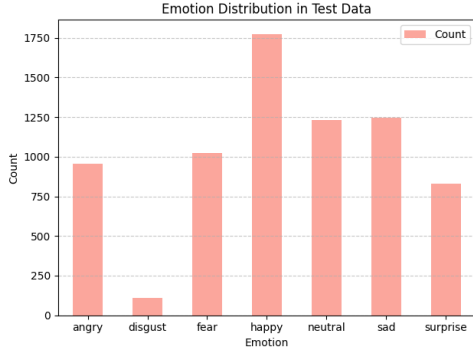


Figure 1.3: Test data distribution.

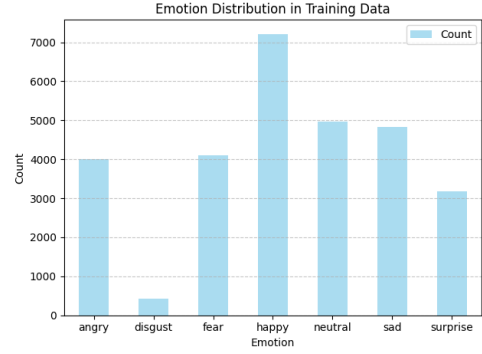


Figure 1.4: Train data distribution.

'disgust', which has the fewest. This imbalance can bias models towards the majority classes and negatively impact performance on minority classes. This can be clearly seen in Figures 1.3 and 1.4 [7].

In addition, FER-2013 contains a degree of label noise, caused by the automated nature of data collection as well as subjectivity in labelling. This makes some images prone to misclassification.

Furthermore, the 48x48 pixel resolution is relatively low, which makes it more difficult for models to capture subtle facial features, which are important for distinguishing between some emotions.

Finally, the dataset provides emotion labels and images but it lacks additional context, such as: age, gender, ethnicity, or the situation in which the expression was captured. This limits the scope of analysis and introduces potential biases.

Researchers have proposed several strategies to mitigate above-mentioned challenges. Data augmentation, usage of weighted loss functions, more robust network architectures and label refinement - all these methods contribute to improving performance on this dataset.

In the Table 1.3, we can see the results of the 2013 ICML contest, for which FER-2013 was created. It is worth noting, that Ian Goodfellow found that human accuracy on FER-2013 was $65 \pm 5\%$ [7].

Team	Accuracy
RBM	71.162%
Unsupervised	69.267%
Maxim Milakov	68.821%
Radu + Marius + Cristi	67.484%

Table 1.3: Private test set accuracy on FER-13

1.3 RAF-DB

The RAF-DB (Real-world Affective Faces Database) is a widely used dataset for facial expression recognition in the wild. It contains approximately 30,000 facial images collected from the Internet, all with diverse real-world conditions such as varying head poses, occlusions, and illumination levels. The dataset is split into two versions: the single-label version (RAF-DB v1.0), where each image is annotated with one dominant emotion, and the multi-label version (RAF-DB v2.0), where images can be tagged with multiple emotion labels to reflect ambiguity in facial expressions.

Emotion labels in RAF-DB are based on Ekman's six basic emotions: happiness, sadness, anger, fear, disgust, and surprise, plus neutral. These annotations were crowd-sourced from about 40 independent taggers per image to ensure robustness and consistency, followed by a screening process that eliminated unreliable annotations.

Figure 1.5 shows example images from RAF-DB v1.0 and v2.0 illustrating the range of expressions and image conditions.

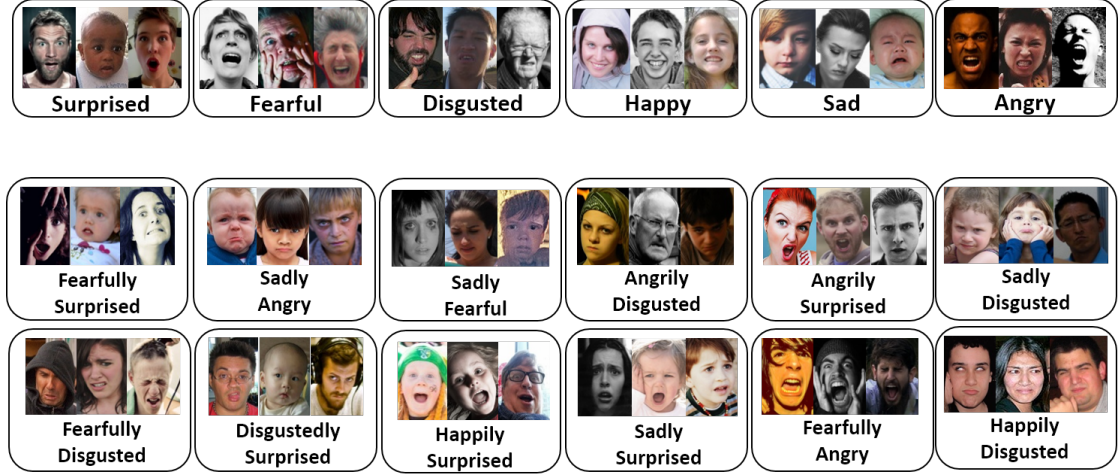


Figure 1.5: Sample images from the RAF-DB dataset, illustrating distinct emotion categories [5].

One of the advantages of RAF-DB is its emphasis on real-world variability, making it particularly valuable for training models that generalize well to uncontrolled settings. However, a limitation is its relatively small size compared to datasets like AffectNet, and the lack of continuous (valence-arousal) annotations.

Figure 1.6 shows data distribution for Gender attribute.

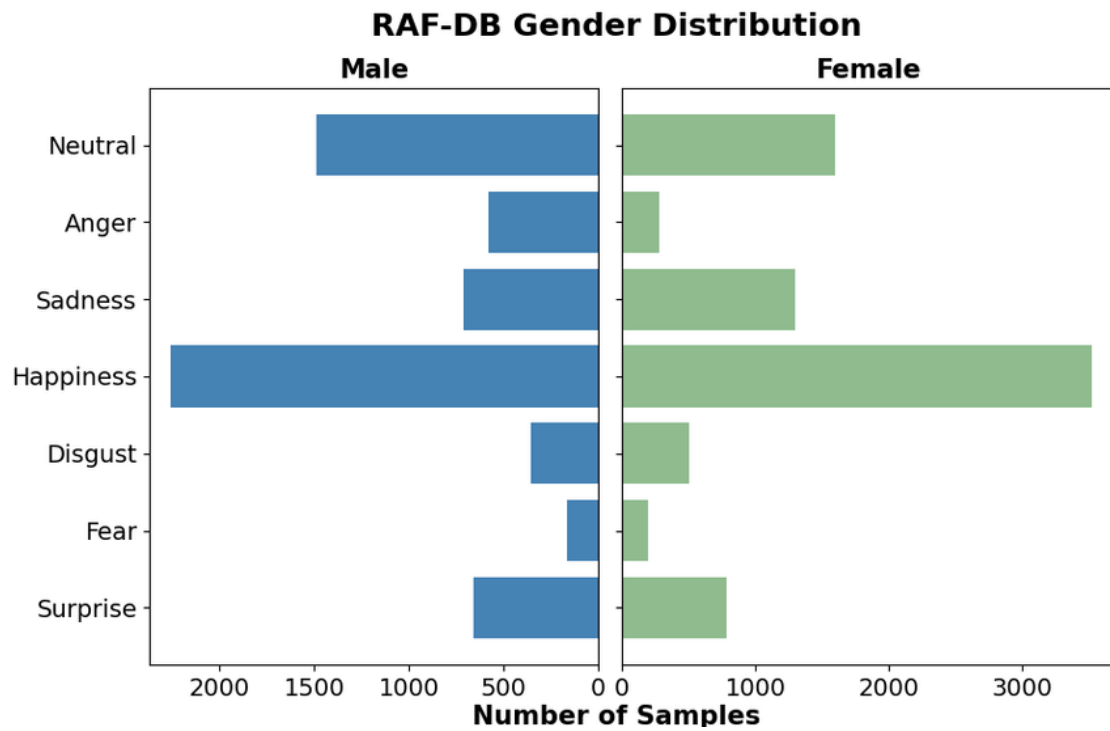


Figure 1.6: Here one may notice that the most disbalanced class is anger, with about 4 times more male images [6].

1.4 Remarks and Conclusions

In conclusion, the three datasets are made to suit different needs. AffectNet with its considerable size and different labeling technics is great for large scale research and precise emotion recognition models. Another advantage of it, is that it was sampled with 6 different languages, giving it a wider range of emotion displays from different cultures each of whom may display an emotion in a slightly different manner. An important aspect of AffectNet is that aside from rotations, which are applied to some images, no other significant transformations were applied to the dataset. One can argue that the pictures are usually taken from different positions, making the dataset split between pictures from the side and direct ones. Nevertheless, the parts of faces that define emotions, example of such would be smile for happiness, are still visible and are, generally speaking, located around the center of images.

RAF-DB posses around 30,000 Internet-sourced images. The images are labeled with 7 basic and 12 compound emotion categories. In case of RAF-DB, there is a great variability in surroundings of the photographs, as well as in demographics. This dataset is also subjected to powerfull transformations. They are directed at blurring the backgrounds, as well as changing the shapes of faces, in many cases. This creates even greater variability, and consequently, the models, trained on it do not achive high testing accuracy on AffectNet and vice versa. This can be explained by the fact that in many cases, RAF-DB classes do not have some common parts of faces that would be similar at the majority of images.

FER-2013 is a dataset with around 40,000 pictures for training and testing. It takes to account 7 basic emotion types. Even though it poses limited resolution and modality, it is lightweight, and thus suitable for developing model prototypes. Unlike AffectNet and RAF-DB, FER-2013 has greyscale images, although this may create some obstructions at detecting some facial features, as red lips that smile, generally, models that were trained on it are capable of detecting said features with relatively high accuracy. Training is also made easier, by making images more focused on the faces, which frees models from processing unnecessary parts of the images.

Chapter 2

Models for Emotion Recognition

2.1 ResNet

After researchers realised the potential of Deep Learning to solve complex problems and learn even more complicated patterns, the pursuit of deeper architectures became a top priority of researchers. However, as neural networks grew in size and depth, the degradation problem became more and more concerning. Unlike overfitting, the degradation problem is correlated with the vanishing gradient problem, where the gradients become infinitely small, obstructing the ability of the first layers of the CNN to learn properly.

In 2015, the research team from Microsoft Research Asia introduced a new architecture of CNNs. Deep Residual Networks (or ResNet) addressed the problem of vanishing gradients. This architecture has won the 2015 ImageNet Large Scale Visual Recognition Challenge (ILSVRC), with a network having more than 150 layers [19].

The core idea of ResNet is residual learning, which is achieved using skip connections (also known as shortcut connections). In a traditional neural network segment (or block), the layers are expected to learn some mapping, called, for example, $H(x)$. In this case, the degradation problem makes it extremely difficult for stacked non-linear layers to learn a mapping $H(x) = x$ (identity mapping), whenever this is the optimal function. As mentioned before, ResNet introduces skip connections. This changes the underlying function that the layer must learn to $F(x) = H(x) - x$. Therefore, the output function of the layer is $Y = F(x) + x$. If the optimal solution for $H(x)$ is indeed an identity function, then the layer output should not change much, which makes $F(x) \approx 0$. This function is much easier for non-linear stacked layers to learn than $H(x) = x$.

There are three most obvious benefits to using residual learning. The first benefit is the mitigation of the vanishing gradient. The gradient can freely propagate

back using the skip connections. The second benefit is the ease of optimisation. This is because it is easier for an optimiser to learn small perturbations from an identity mapping (when $F(x) = 0$) than it is to learn the whole identity mapping $H(x) = x$ from scratch. The third benefit is the ability to construct deeper architectures with more layers. Since the degradation problem is dealt with, we can concentrate on creating more complex models with sometimes even hundreds of layers.

2.2 Visual Geometry Group

Researchers started looking into how network depth might affect performance as Convolutional Neural Networks became more and more popular in computer vision. Before 2014, the majority of CNN designs included only a few convolutional layers and were therefore somewhat shallow. By showing that much deeper networks might perform better, the Visual Geometry Group (VGG) at the University of Oxford changed this norm. They made a new VGG model family, of which VGG-16 and VGG-19 are the most well-known models.

The VGG architecture was built with a simple idea in mind: using very small (3×3) convolutional filters consistently throughout the network while dramatically increasing depth. This approach contrasted with earlier architectures like AlexNet, which used larger filters of varying sizes ($11 \times 11, 5 \times 5, 3 \times 3$).

VGG-16, the version that is used the most, is made of 16 layers (13 convolutional and 3 fully connected) with around 138 million parameters. The network processes input images of 224×224 pixels and follows a systematic design where the number of channels doubles after each max-pooling operation ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 512$).

VGG's main contribution was showing how depth affects CNN performance. By increasing the network depth from 11 to 19 layers, the researchers showed consistent accuracy gains on the ImageNet dataset. VGG-16 outperformed previous models with a top-5 error rate of 7.3 on ImageNet. In addition to laying the groundwork for even deeper networks like ResNet and DenseNet, this work established depth as a fundamental design principle for CNNs. figure 2.3 shows the simplified VGG-16 architecture.

The VGG models showed that remarkable outcomes could be obtained with a simple architecture and depth, which made them excellent feature extractors for transfer learning applications in a variety of computer vision tasks in addition to being strong classifiers.

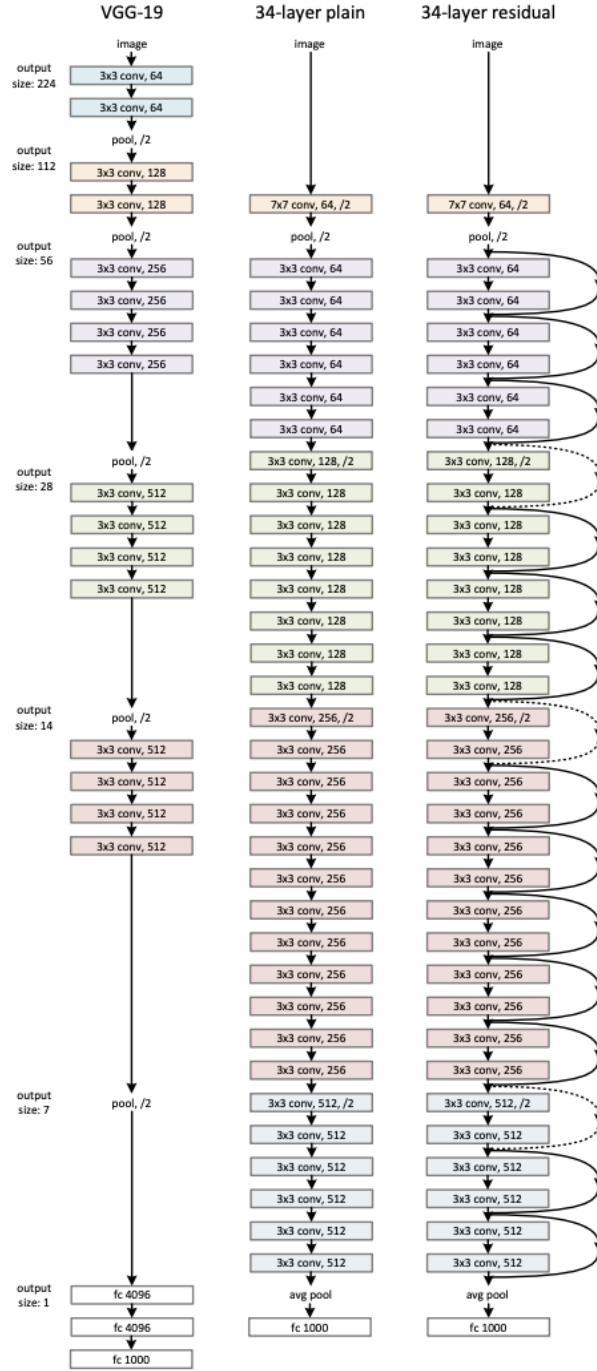


Figure 2.1: Architectures used in ILSVRC2015 [8]

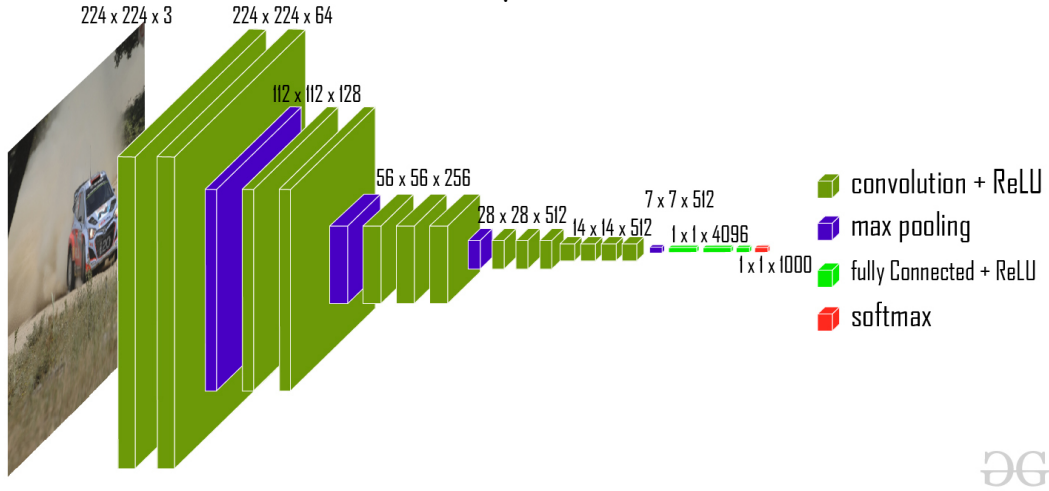


Figure 2.2: VGG-16 architecture [10].

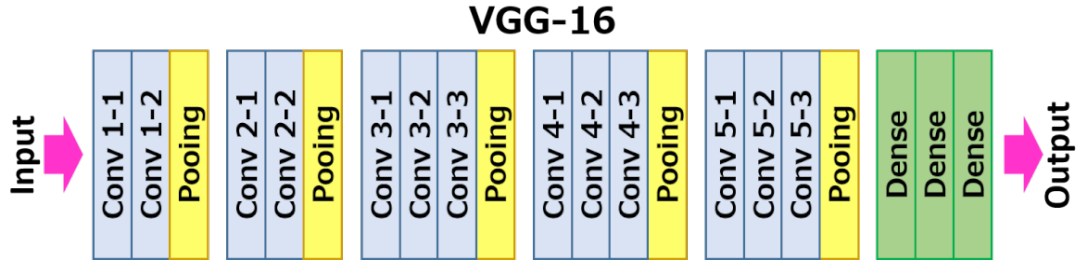


Figure 2.3: VGG-16 simplified architecture [11].

2.3 EfficientNetB0

As Convolutional Neural Networks changed with time, research teams began focusing on new models that are more computationally efficient, besides being more accurate. Deeper and wider networks can provide better performance, but these improvements often require significantly higher computational power. To address this issue, researchers at Google devised a new family of CNN models known as EfficientNet. The baseline model in this family is EfficientNetB0, which serves as the foundation for all following, larger variants.

EfficientNetB0 was developed using Neural Architecture Search (NAS), a technique that automatically discovers high-performing network architectures. It introduced an innovative compound scaling method to systematically scale up the models depth (number of layers), width (number of channels), and resolution (input image size) in a balanced manner. This scaling approach differs from prior methods that typically scaled only one dimension at a time, often resulting in sub-

optimal performance or excessive resource usage. Figure bellow shows different ways of model scaling.

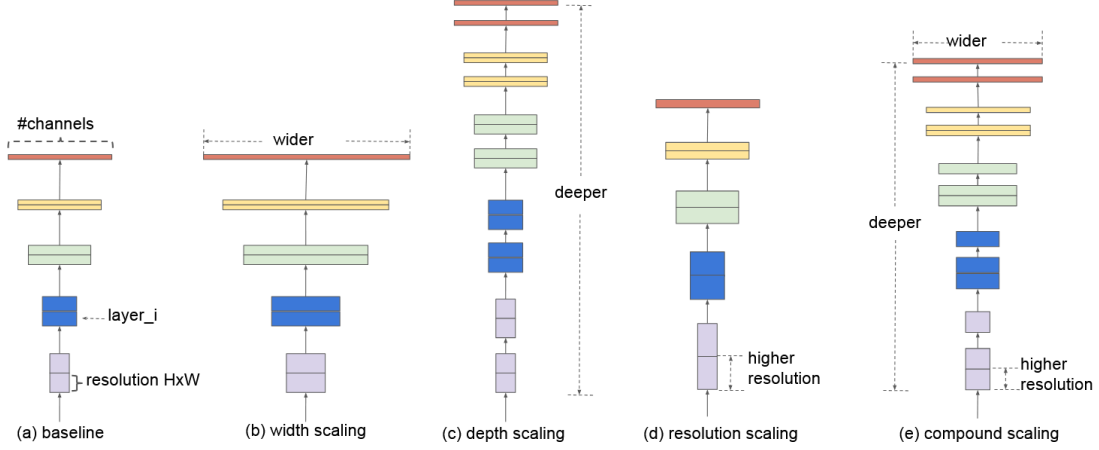


Figure 2.4: Model Scaling

The B0 model itself is relatively lightweight, with approximately 5.3 million parameters and an input image resolution of 224x224 pixels. Despite its modest size, EfficientNetB0 achieves competitive accuracy on benchmark datasets like ImageNet. By serving as a carefully-optimized starting point, it allows the creation of larger models (EfficientNetB1 through B7) that progressively increase in accuracy and complexity while still maintaining computational efficiency.

EfficientNet's main contribution is its compound scaling method that scales CNN depth, width, and resolution simultaneously. This approach achieves state-of-the-art performance with significantly fewer parameters than existing models, making it practical for resource-constrained deployments. The framework shows that principled scaling outperforms naive approaches that only increasing model size.

2.4 ChatGPT

ChatGPT is capable of analyzing emotions in two "dimensions". It is capable of distinguishing emotions embedded into the queries it is given via speech or text. It is also capable of "understanding" emotions of people, whose faces' photos were submitted to it. Bellow is information about both aspects of emotion recognition by ChatGPT. A work by Omar Banimelchem and Wlla Amayreh - "The Performance of ChatGPT in Emotion Classification" provides an insight on how well does ChatGPT classify emotions in text. The graph 2.6 shows the confusion matrix of these predictions.

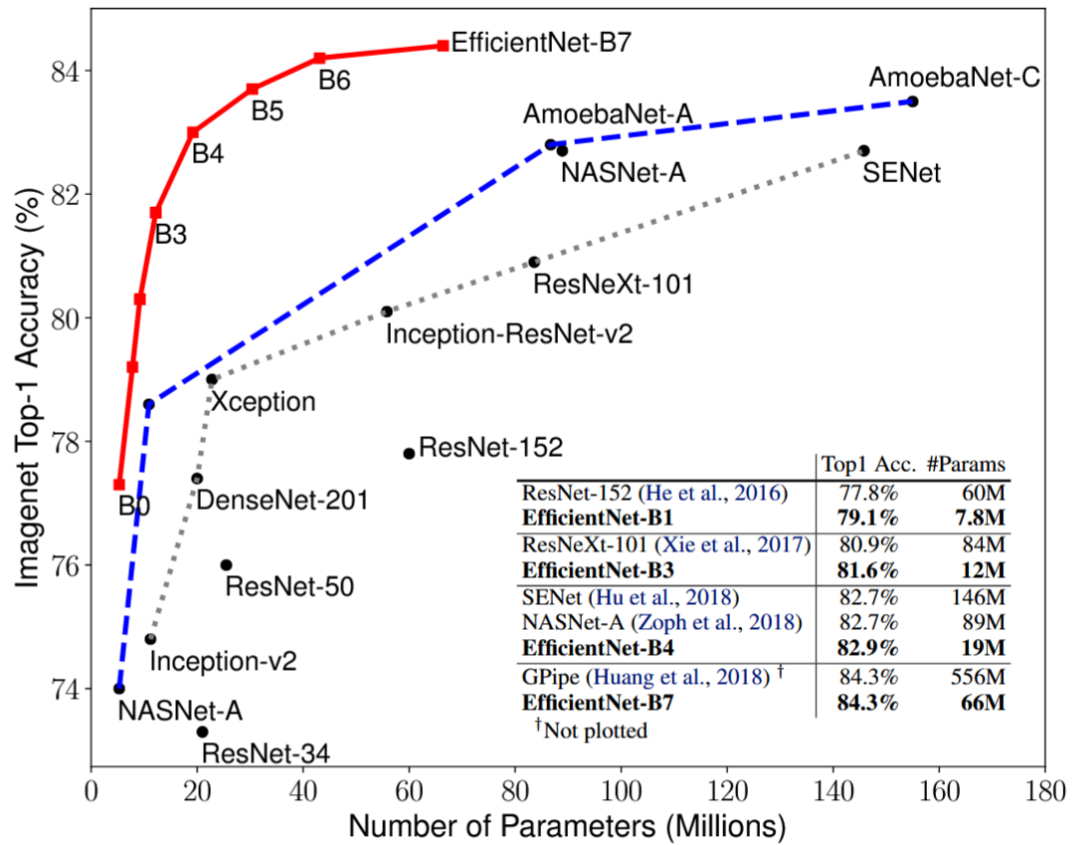


Figure 2.5: Model Size vs. ImageNet Accuracy [19]

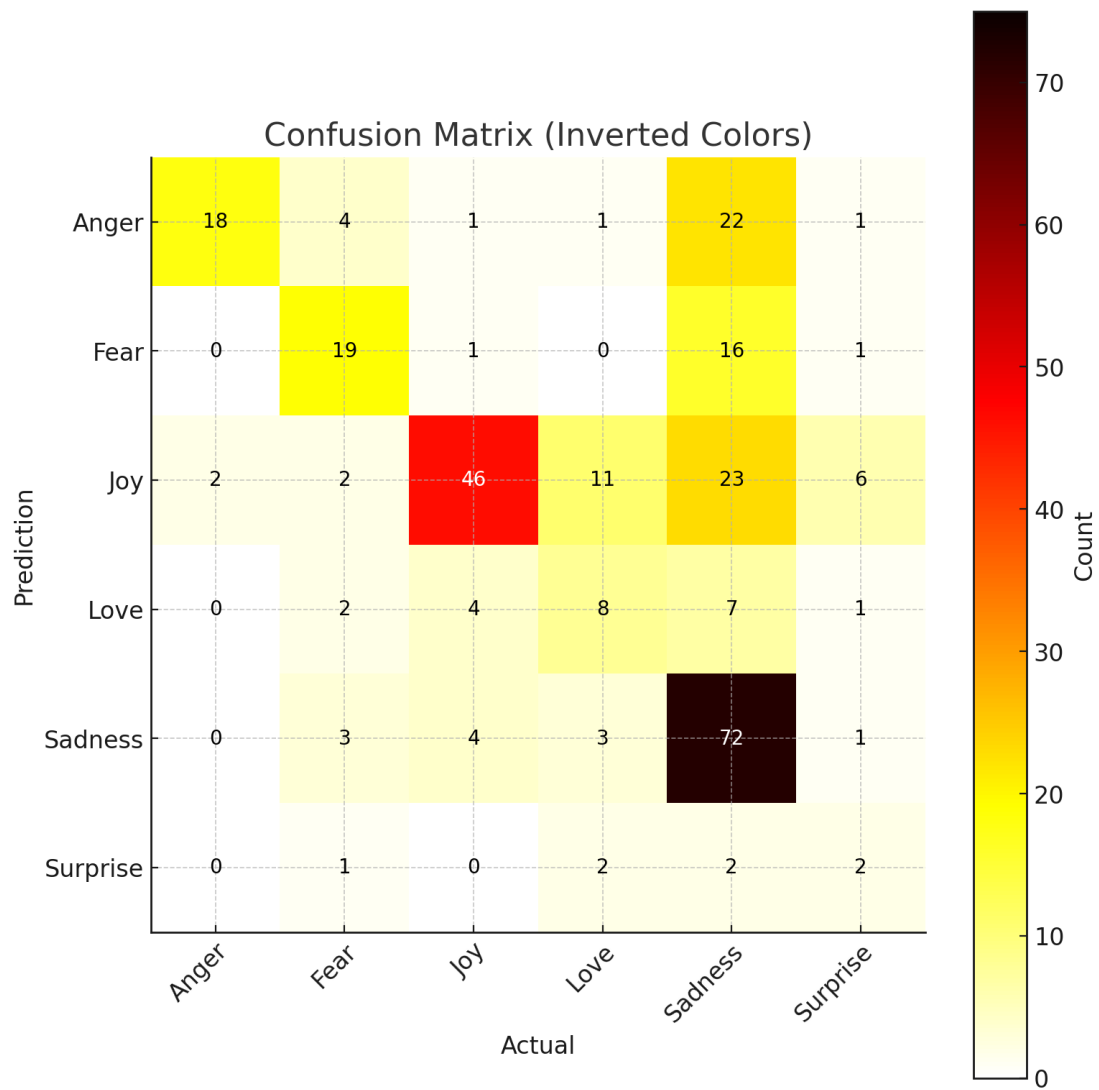


Figure 2.6: Confusion matrix for ChatGPT query emotion recognition [12].

ChatGPT processes images differently, than ResNet. It does not apply kernel filters to images, instead it splits images into sub matrices, flattens them out into vectors, and then operates on them. The study "Identifying Basic Emotions and Action Units from Facial Photographs with ChatGPT" by Robin S. S. Kramer explores its accuracy. The table bellow shows the testing results.

Emotion	No. ^a	H. Ag. (%) ^b	FaceReader (% Cor.) ^c	ChatGPT (% Correct)	
				7AFC	9AFC
Anger	30	86.5 (9.7)	76.7	100	100
Disgust	30	90.6 (6.2)	92.9	100	100
Fear	30	68.9 (13.1)	87.5	86.7	83.3
Joy	30	87.6 (8.4)	96.7	100	100
Neutral	30	67.0 (13.3)	93.3	100	100
Sadness	30	86.9 (11.7)	76.7	100	96.7
Surprise	30	89.0 (6.4)	92.6	100	100
Overall	210	82.4 (13.7)	87.9	98.1	97.1

^a Number of images.

^b Human agreement score is shown with standard deviation in parentheses.

^c FaceReader 6 software.

Table 2.1: A summary of model performance on the emotion categorisation task. [13]

2.5 Remarks and Conclusions

In conclusion, the models described in this chapter serve the following purposes. ResNet is a well-established solution that is used for various reasons. It is integrated in python libraries, such as torchvision. That makes it very ubiquitous in education as well as other fields. The idea that allowed ResNet to be as large as 150 layers is the integration of the skip-layers. The problem of the vanishing gradient is solved in this way. Secondly, VGG architecture introduced the approach of using only small convolutional filters, which optimized the model. Combining depth and efficiency, VGG-based models achieve high accuracy. EfficientNetB0 is focusing on achieving efficient solutions. It is relatively lightweight and works for image resolution 224x224. The EfficientNet framework allows designing CNNs for embedded and mobile devices. ChatGPT can process emotions as text and as images of the expressions. The analysis of the images is unorthodox, using no kernels, and instead operating on vectors.

Chapter 3

Metrics for Evaluation

3.1 Accuracy

This metric represents the ratio between the correct and false predictions. The Formula (3.1) shows the method for calculating the accuracy in the realm of multiple classes. It is derived from the total confusion matrix [16]:

$$Accuracy = \frac{\sum_{i=1}^N M_{i,i}}{\sum_{i=1}^N \sum_{j=1}^N M_{i,j}} \quad (3.1)$$

where M denotes the confusion matrix, $M_{i,j}$ is the number of samples of class i predicted as class j , $M_{i,i}$ represents correctly classified samples for class i , and N is the total number of classes.

Although the perfect precision is 100%, training different emotion recognition models on different datasets proved that the accuracy can vary significantly from models, training sets, and testing sets, as well as Python libraries used. The go-to accuracy for our training was 70% validation accuracy. It was achieved on different models, trained on AffectNet. Fer-2013 and RAF-DB had lower testing accuracy of around 65%. It is worth noting that 2 approaches were used. First was to split the datasets in 2 sets: training and validation. The testing would be performed on the other datasets, for instance, training and validation could be done on AffectNet, and the testing - on RAF-DB. This approach was changed to the one, of splitting data set into 3 parts, usually with 14:3:3 or 8:1:1 ratio with training, validation, and testing subsets.

Figure 3.1 shows the testing accuracy of different models used. The models were tested on other datasets after training as well. The highest accuracy in this case was shown by tests on CK+ dataset. While others were averaging around 40-50%, it scored around 76% accuracy on a model, trained on ResNet50 and 71% on

ResNet18. The fact that in the most cases accuracy of testing subsets of datasets, on whose training subsets models were trained showed the better accuracy, goes to show that in each dataset there are certain patterns which are unique to it, which makes its accuracy on other datasets significantly lower. Such patterns can be strong image filtering, as in case of RAF-DB, or the fact that dataset is too small, as in JAFFE, and thus, model tries to generalize features of a few people to thousands.

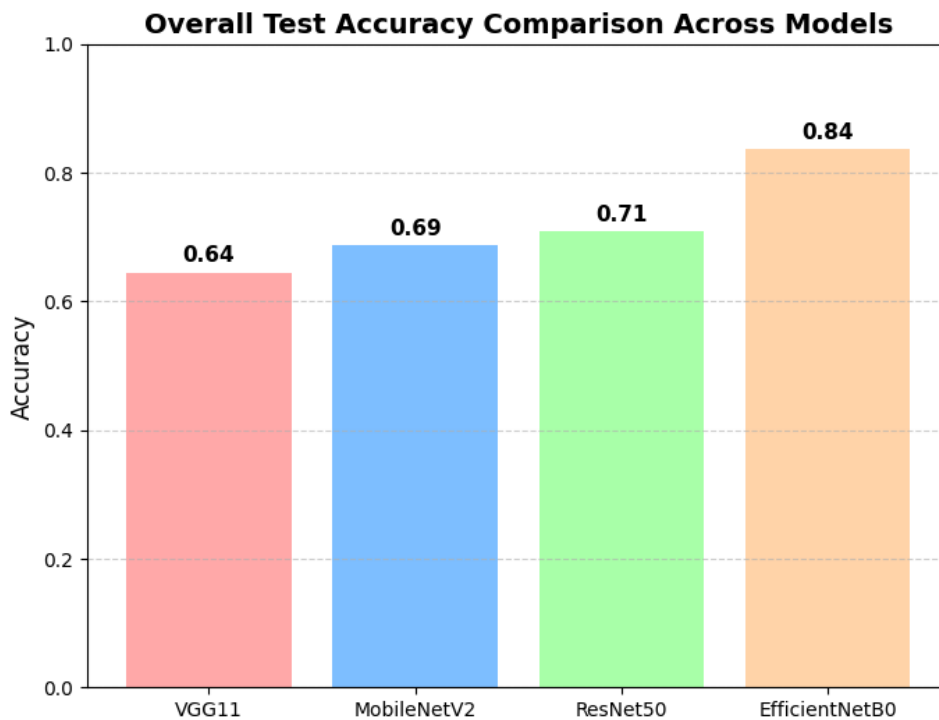


Figure 3.1: Testing accuracies of selected models for different datasets.

3.2 Recall, Precision, F1-score

Recall, Precision and F1-score are important metrics of model performance. They are defined and most easily understood in binary classification problems. Since we have a multi-class classification problem, we have to use one of the strategies to divide multi-class classification into several binary classification problems.

For our purposes, we shall use the OvR, or one-versus-rest approach. The method is as follows: for each class i of N classes we have, we introduce a binary classifier which distinguishes i -th class as positive, and all other classes as negative.

Figure 3.2 shows the confusion matrix definition for a binary case of classification problems. In order to calculate recall and precision, it is required to calculate the confusion matrices for each class, using the OvR method mentioned before. Once we have all those 2×2 matrices, we can apply the respective formulas for precision and recall, to calculate per-class statistics. Finally, we can calculate the overall recall and precision by applying either weighted or macro- average.

		PREDICTED	
		NEGATIVE	POSITIVE
ACTUAL	NEGATIVE	TN Count of TN	FP Count of FP
	POSITIVE	FN Count of FN	TP Count of TP

Figure 3.2: Confusion matrix for a binary classification problem.

Recall measures the proportion of actual positive cases that our model correctly identified. In Formula (3.2) you can see how we can calculate the metric, where TP is True Positive and FN is False Negative [16]:

$$Recall = \frac{TP}{TP + FN} \quad (3.2)$$

Precision measures the proportion of positive predictions that were actually correct. In Formula (3.3) you can see how we can calculate the metric, where FP

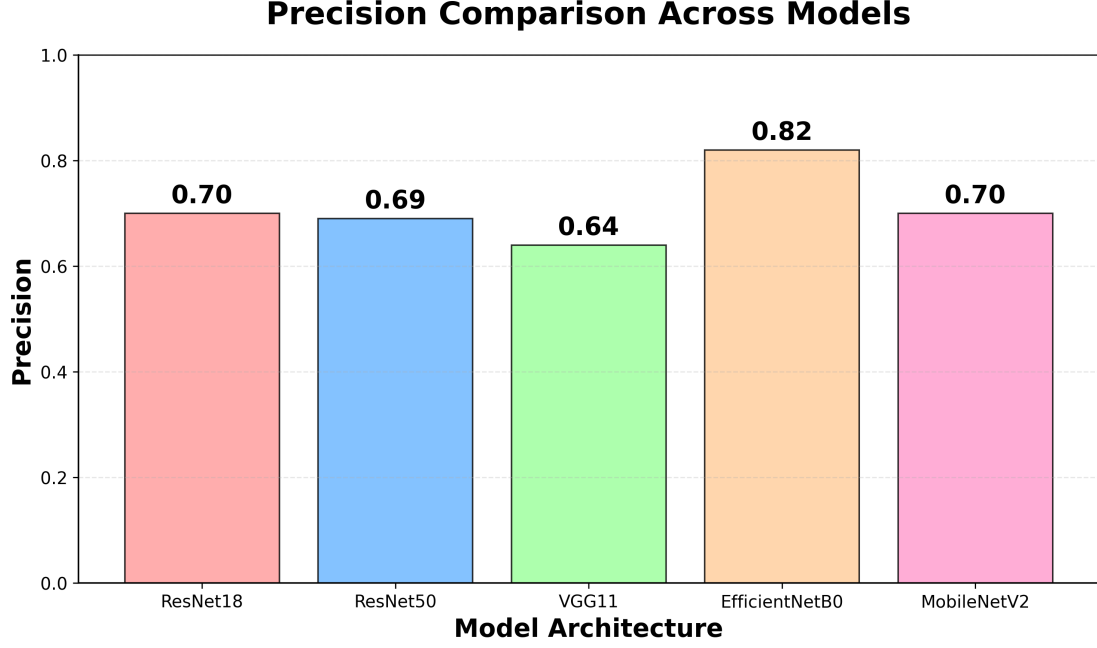


Figure 3.3: Graph of macro-average precision of our models.

is False Positive [16]:

$$Precision = \frac{TP}{TP + FP} \quad (3.3)$$

The F1-score is the harmonic mean of Precision and Recall. It provides a single metric that balances both recall and precision. Formula (3.4) shows how to calculate the metric [16]:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (3.4)$$

In Figures 3.3 and 3.4 we can see the macro scores of recall, precision and F1 across different models. It is worth noting that VGG11 and ResNet50 were trained on datasets AffectNet, FER-2013, NHFIER and RAF-DB. We believe that RAF-DB is a peculiar dataset (because of how images are preprocessed there), hence the statistics of the models are lower than for those models where RAF-DB was not used during training.

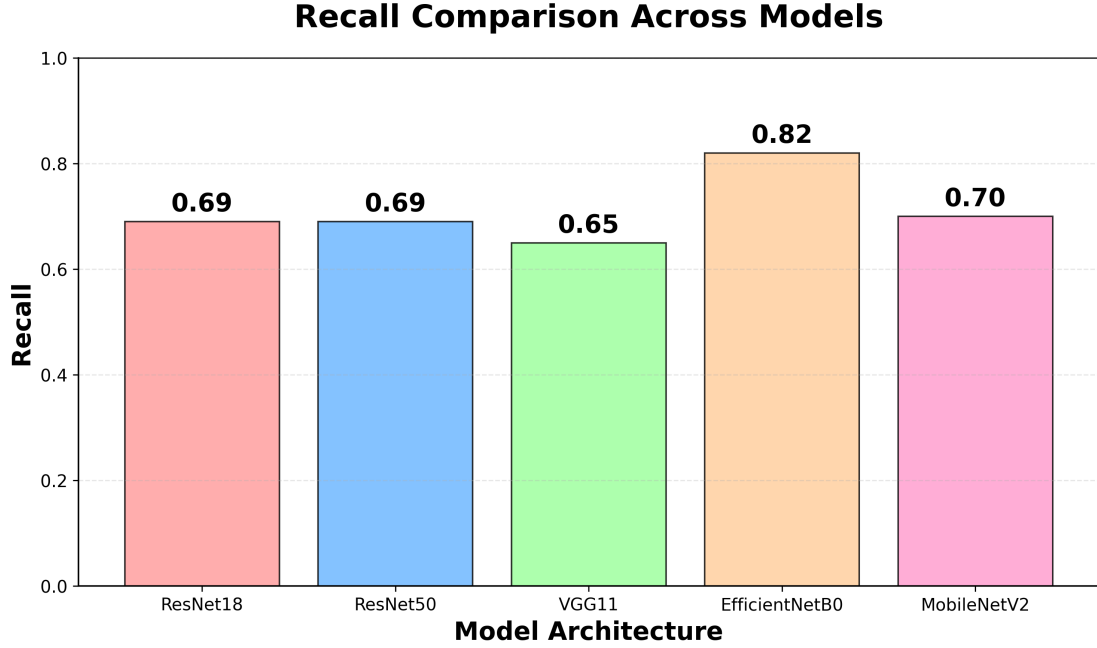


Figure 3.4: Graph of macro-average recall of our models.

3.3 Area Under Curve called Receiver Operating Characteristic

The AUC score can be interpreted as the probability that the model will rank a randomly chosen positive instance higher (assign a higher predicted score) than a randomly chosen negative instance. While generally AUC is a metric for models for binary classification, we can generalize this statistics using the OVR (one-versus-all) approach. The approach is to break down our classification task into multiple binary problems, where one class is designated the "positive" class, and all others are grouped into the "negative" class. Then, after calculating the AUC for each class, we can find the macro-average for our model.

In Table 3.1, we can see the AUC scores of our models, tested on the test set of the training data. It is worth noting that models ResNet50 and VGG11 were trained on datasets AffectNet, FER-2013, NMFIER and RAF-DB, however their AUC score is not much lower than that of other models.

In Figure 3.5, we can see the AUC curves of the selected models plotted. As we can see, EfficientNetB0 gives us the best score, which makes this model one of the top candidates as our model of recommendation for emotion recognition.

Model Architecture	AUC-ROC Score
ResNet18	0.95
ResNet50	0.92
VGG11	0.91
EfficientNetB0	0.98
MobileNetV2	0.95

Table 3.1: AUC-ROC Scores of Model Architectures.

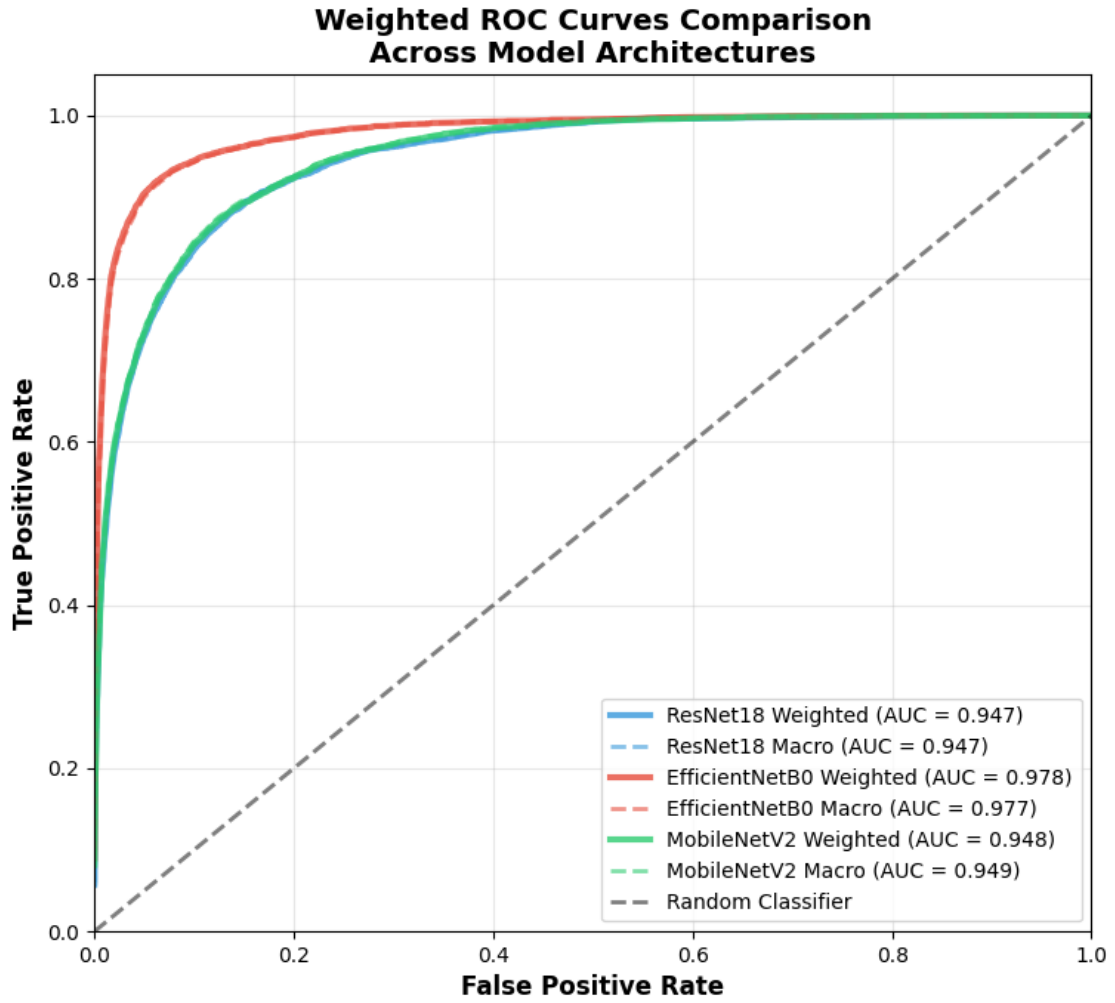


Figure 3.5: AUC curves of selected models

3.4 RAM Capacity and Training Time

The training periods depend highly on the following factors. Above all else is the size of the training dataset. This can be traced to the training of ResNet50 on both undersampled balanced datasets and oversampled datasets, which had been expanded, so that all emotion class sizes are equal to the largest class. Another big factor was the model. While ResNet50 took around 8 hours to train on joint oversampled dataset of AffectNet, FER-2013, RAF-DB and NHFIER or JAFFE, with total of around 100'000 images, VGG11 took 13 hours, and have not even reached the same accuracy. If a model was trained on a smaller dataset, like in the case of NHFIER (Natural Human Face Images for Emotion Recognition), which had 5600 images, the training time was significantly lower, in this case - 25 minutes per 25 epochs. The EfficientNetB0 trained on joint oversampled dataset of AffectNet, RAF-DB, and NHFIER took about 10 hours for 10 epochs.

The computer resources employed for training have been different, since training has mainly been conducted on 2 machines - Asus laptop with Intel i5 processor and 16 GB of RAM as well as 4 GB of VRAM on its GPU, and MacBook M3 Pro, with 16GB of unified RAM. In the case of first machine, the RAM used was constant - 0.5/1.5 GB of RAM and 3.5 GB of VRAM. In the case of the second, memory usage fluctuated around 800 MB. It is also worth noting, that Training on MacBook of ResNet18 and MobileNetV2 took about 2 hours, which is significantly faster than the same model training time on Asus laptop. This might be explained by the fact that training set on the MacBook M3 Pro did not include RAF-DB, whose emotion expressions are generally different from the rest of the datasets. The testing time in most cases was less than a minute.

In the Table 3.2 one can see the training time per model on different computers.

Table 3.2: Training times of different models across 2 computers with different architectures.

Model	Computer Setup	Training Time (hours)
VGG11	ASUS (with RAF-DB)	13
ResNet50	ASUS (with RAF-DB)	8
EfficientNetB0	MacBook (no RAF-DB)	5
MobileNetV2	MacBook (no RAF-DB)	2
ResNet18	MacBook (no RAF-DB)	4

3.5 Remarks and Conclusions

Having done the trainings and testings of the models with different architectures, datasets, libraries, approaches, and machines we have come to the following conclusions. First of all, the better library, in our humble opinion, for training models and achieving higher accuracies is Pytorch. While we have tried to use both Pytorch and Tensorflow, the first showed significantly better result metrics. However, it is worth noting, that Tensorflow provides its users with a less steep learning curve, as well as a great documentation for studying it from the beginning. Nevertheless, the Pytorch was used, since it provided a jump of 20% in accuracy, compared to Tensorflow, even when pretrained Imagenet weights were used.

The second aspect that is worth mentioning is that different datasets are very unique and one must take that into consideration, while adjusting model training parameters, such as number of epochs, number of dropout layers, as well as learning rate decay function and optimizers. The datasets with high variability of data and preprocessing were harder to learn for the models on. The same applies to testing on such datasets, as well as to testing on small datasets that had not enough samples to let a model use everything it learned, while also having features, which it has not learned.

The third thing we analyzed during the model training, is how models of different sizes performed. The interesting thing here, in our opinion, is that too large models cannot be used for the following reason. If the model has too many parameters, it will be capable to learn to recognize the training set, without learning the patterns of different emotions that actually make model recognize the emotions on different testing sets. We discovered that no matter which model we used, it either severely underfitted, as in the case of Tensorflow implementations, or eventually overfitted, when training accuracy reached around 100%, while validation accuracy was only about 65-70%.

To summarize, our solutions achieved a descent level of performance, if one considers above 90% macro and weighted ROC metric, which is the most significant metric in the real-world applications. Table 3.3 shows the weighted AUC.

VGG11	ResNet50	MobileNetV2	EfficientNetB0	ResNet18
0.93	0.97	0.95	0.98	0.95

Table 3.3: Weighted Area Under Curve (AUC) for different models.

Chapter 4

Documentation from Group Project

4.1 System Architecture

The emotion recognition application consists of two main, and five side components. Figure 4.1 demonstrates it. The Frontend module is the part, with which Educators and Users can interact. It consists of two parts: Dashboard Interface and User Interface. Dashboard interface provides Educators with the emotion statistics of the Students, while the User Interface provides Users with customizing menu for the model training. Backend consists of three parts: Web Server, Custom Module Training, and Student's side(Emotion Analysis Module). The last is concerned with taking pictures of Students and processing them, after which it sends the information to the Frontend. The Custom Model Training submodule is responsible for training the model, which is customized by the User. Web Server is the thing that connects Backend to Frontend, having API.

User Interface Module provides Users with emotion recognition model training capabilities. The main part of it is the menu, where it is possible to set all necessary parameters for the model training. This data is then communicated to Backend module, Custom Model Training submodule, with it carrying out the training. User may also choose displayed statistics of the training, such as ROC, Loss, Precision, F1-score, etc.

It is responsible for presenting comprehensive emotion analytics to teachers. It queries the data of currently running session and presents it in a user-friendly format with interactive visualizations. The dashboard allows users to view emotion distribution charts, track emotional trends over time, compare individual student engagement levels, and generate reports for specific time periods or classes.

This Submodule processes video frames captured from student webcams to extract valuable insights about their emotional states. It uses deep learning-based facial recognition algorithms and convolutional neural networks to identify and

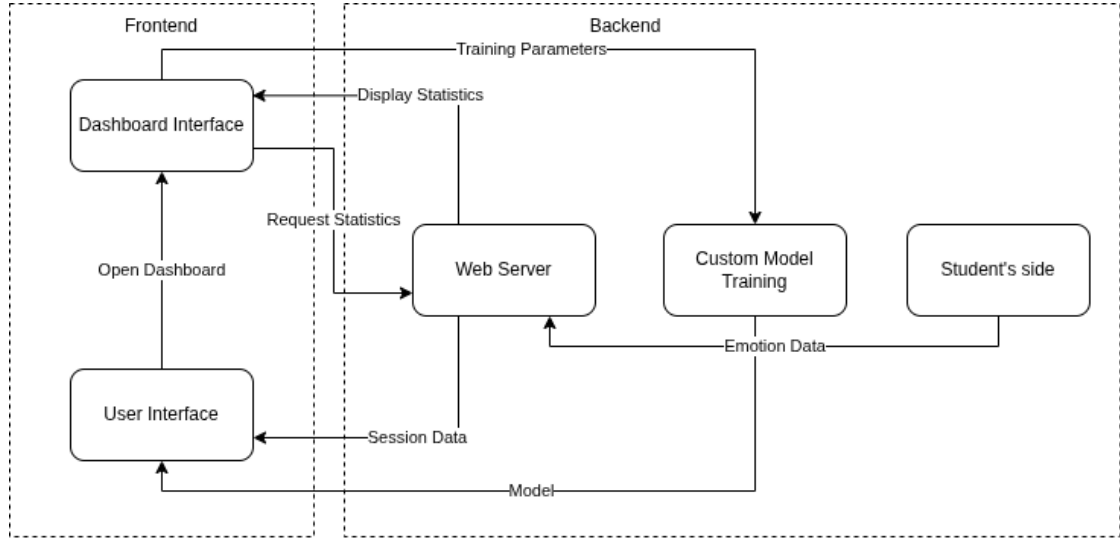


Figure 4.1: Application Architecture Diagram

track key emotional components in the video stream, such as happiness, surprise, sadness, anger, disgust, and neutrality. The module analyzes facial expressions every few seconds, classifies the detected emotions with confidence scores, and stores the extracted data in the memory, which is later used to generate insights and statistics for teachers.

4.2 Communication

The communication between the modules within the application is essential for delivering real-time and accurate emotional data across the frontend, dashboard, and analysis sections. Each module has a distinct responsibility, but they work together seamlessly to create an integrated teaching support experience.

The frontend and backend of the application are developed using Python, utilizing the FastAPI web framework along with HTML, CSS and JS for the frontend interface, which facilitate communication through RESTful API endpoints. When a teacher logs into the system, the frontend sends HTTP requests to the backend API to authenticate the user and retrieve class information. During active sessions, the frontend establishes WebSocket connections for real-time emotion data streaming, allowing teachers to see live updates of student emotions without page refreshes. The backend processes these requests, queries the stored recognized emotion records, and returns JSON-formatted responses to populate the dashboard and monitoring views.

The line of communication is a core part of the application and requires a

complex relationship between the frontend, backend, and processing components to deliver accurate insights. When a class session begins, the frontend initiates video capture from each student's webcam and streams the video data to the backend server. The backend forwards these video frames to the Emotion Analysis Module at regular intervals (every 3-5 seconds). The analysis module processes each frame using pre-trained emotion recognition model, detects faces, extracts facial features, and classifies the emotional state. The recognized emotion along with confidence scores (or we pick emotion wiht highest confidence as detected one) and timestamp are then sent back to the backend, which stores this information in the memory with references to the specific student. The backend simultaneously pushes this real-time emotion data to the frontend via HTTP connections, enabling teachers to see immediate updates on their monitoring dashboard.

When a teacher accesses the dashboard, the backend queries the data of current class session to aggregate emotion data across different time periods. The retrieved data is processed and formatted into statistical summaries, including emotion frequency distributions, average engagement scores, and temporal trends. This processed information is then sent to the frontend as JSON data, where JavaScript renders interactive graphs and charts that help teachers understand student engagement patterns and identify students who may need additional support.

4.3 Main Components Description

The activity diagram shown in Figure 4.2 illustrates the range of actions available within the application, from launching the app to accessing different features.

Figure 4.3 presents the class diagram, which describes how the different parts of the program are related to each other.

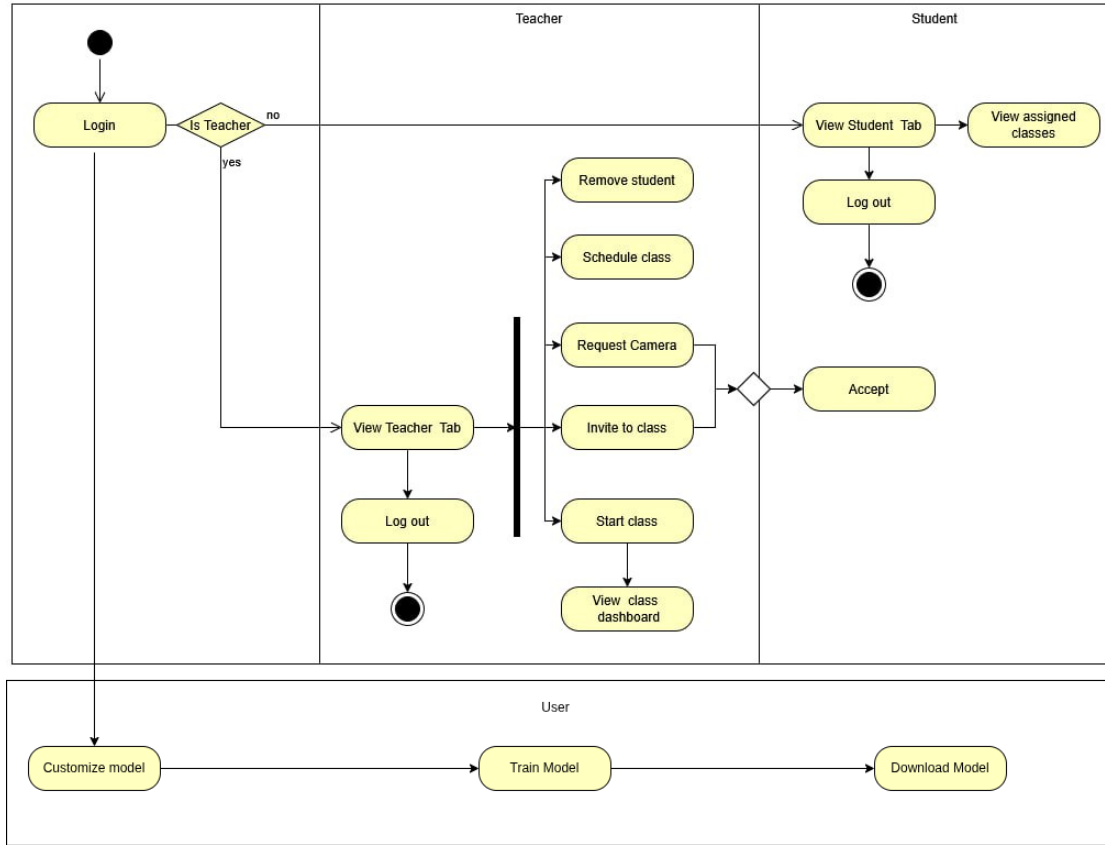


Figure 4.2: Activity Diagram

4.4 Remarks and Conclusions

During the development of the application we have achieved many things that were new to us. The communication between Student and Educator was an insightful thing to work on. Another important thing that was achieved, is that we have created a framework for custom model training. While developing this part of the application, we have thought of the young researchers of the same caliber as ourselves. This does not imply that we did not put much effort into it, on the contrary – we have included many features that we would find helpful ourselves. On the figure 4.4 below one can see a part of the custom model training interface.

The GUI of the application was developed by us in multiple stages. The main accent was on making it user-friendly. It was also important to include such details as main menu and the names of our team members, as well as name and title of our thesis Supervisor. On the figure 4.5 one can see a screenshot of the main menu.

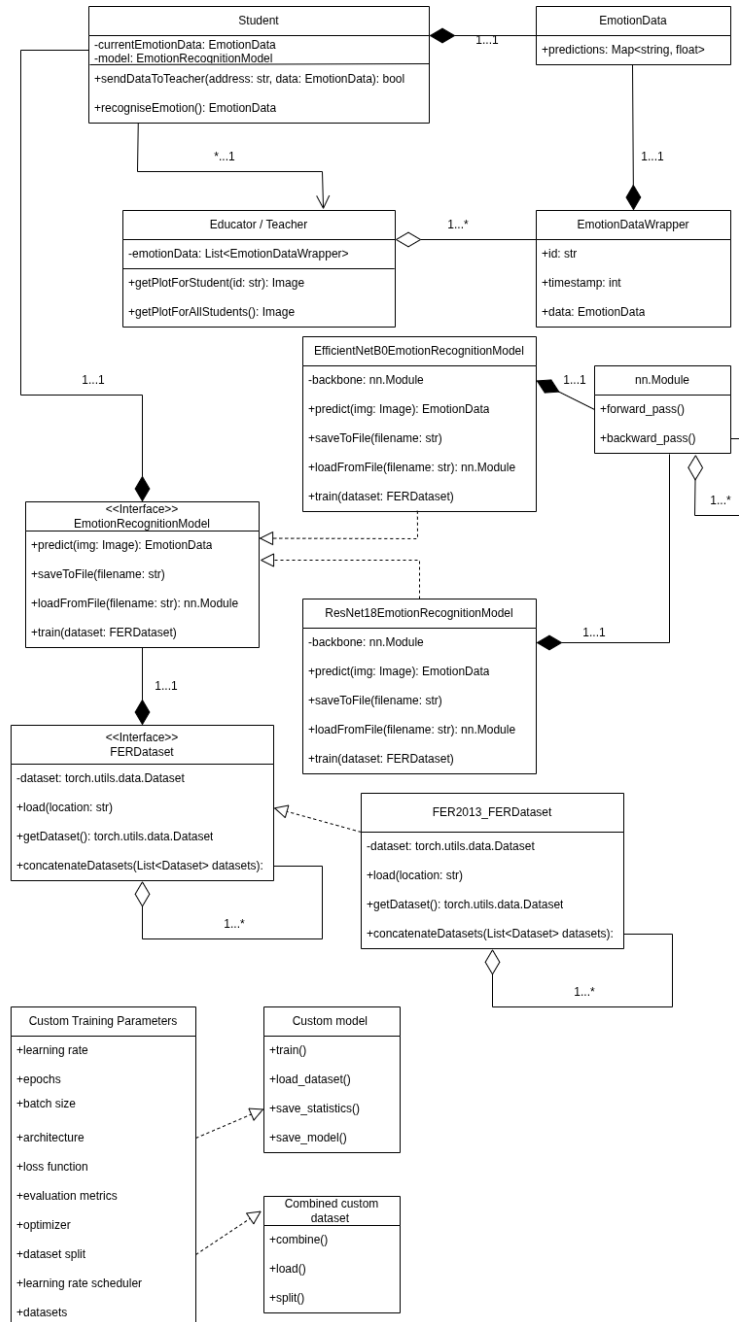


Figure 4.3: Class Diagram

Model Training dashboard

Configure and train your custom emotion recognition model

Select Dataset(s)

☐ AffectNet
 ☐ FER-2013
 ☐ NHFIER

Model Architecture

-- Select Model --

Optimizer

AdamW

Learning Rate Scheduler

CosineAnnealingLR

Loss Function (Criterion)

CrossEntropyLoss

Evaluation Metrics

☒ Accuracy
 ☐ Precision
 ☐ F1-Score
 ☐ ROC AUC

Number of Epochs

10

Figure 4.4: Customizable characteristics of the model trained

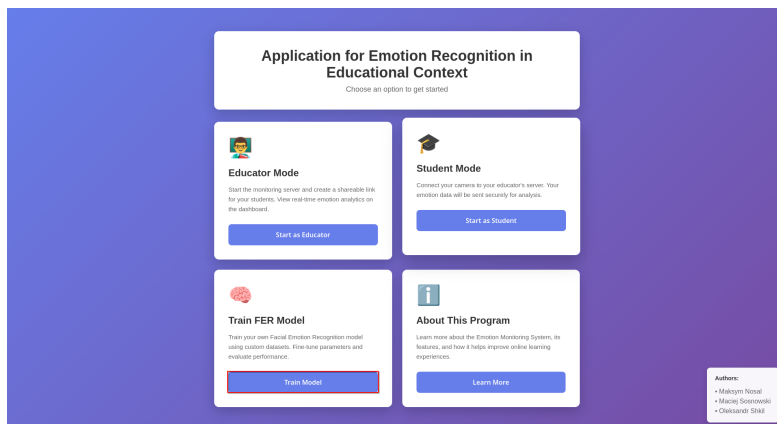


Figure 4.5: The names of creators are at bottom-right corner.

Chapter 5

Experimental Results

5.1 Experiments with Emotion Recognition

We have conducted multiple tests of different models, using testing sets of the datasets. We have been splitting datasets in proportion 14:3:3, as well as 8:1:1 to that. In such a case, the largest part would be the training subset, and the two smaller ones would be the validation set, used to estimate validation accuracy after each training epoch, and the testing subset, which yields model's final statistics. For instance, for the EfficientNetB0 model it is slightly above 70% accuracy on the testing subset. For the ResNet18 it is similar. During training, using pre-trained weights from Imagenet overfitting of the model began at around 60-65%. Not using the weights caused underfitting, with validation accuracy being below 50%. On the figure 5.1 below one can observe validation accuracy after training epochs for the two models: EfficientNetB0 and ResNet18.

Interestingly, while loss gradually decreased with each epoch, validation loss remained at around 1.2. Figure 5.2 demonstrates losses, both regular and validation for both EfficientNetB0 and ResNet18.

Figure 5.3 stating that EfficientNetB0 had better f1 score throughout majority of the epochs reiterates that it is a better model.

However, in real-life applications, one does not simply use the app to see if it is good at recognizing the emotions of already-established datasets. So to test our application in near-to-real-world conditions, we constructed our own dataset - MMO (Maksym, Maciej, Oleksandr), with 8 basic emotions: anger, contempt, disgust, fear, happiness, neutral, sadness, surprise. Each of one took 3 pictures of each emotion; front of the face, left, and right. On the Figure 5.4, one can observe example images from the dataset.

To test the application on our dataset, we decided to not simply run the dataset through the model. This would be optimal to test the model itself, however we

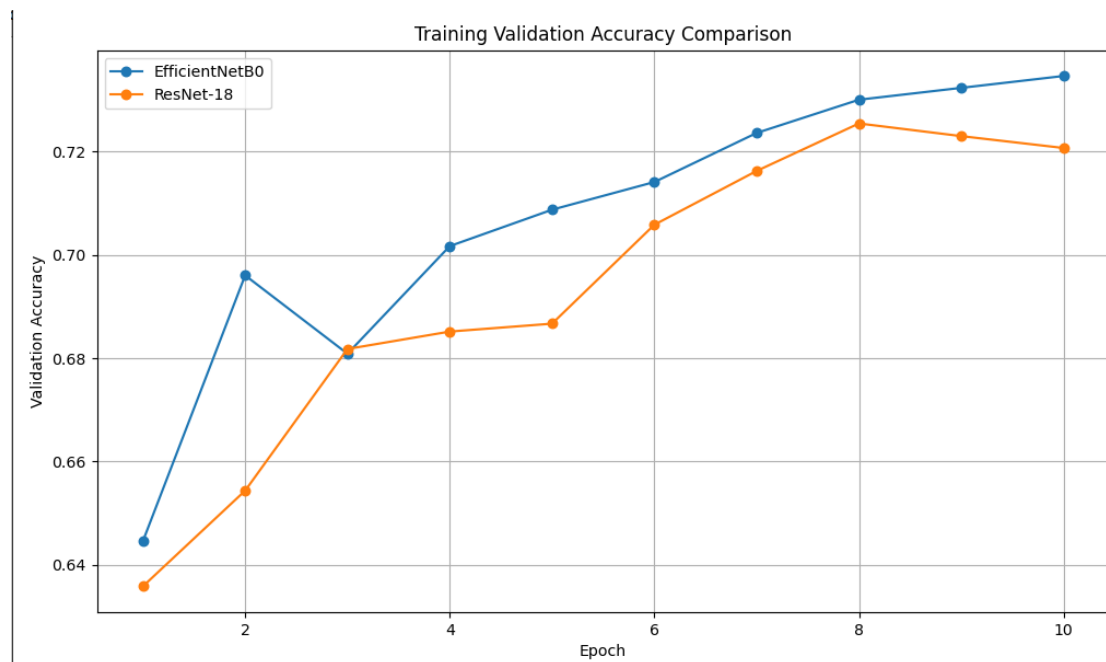


Figure 5.1: EfficientNetB0 has slight upperhand which is nonetheless notable.

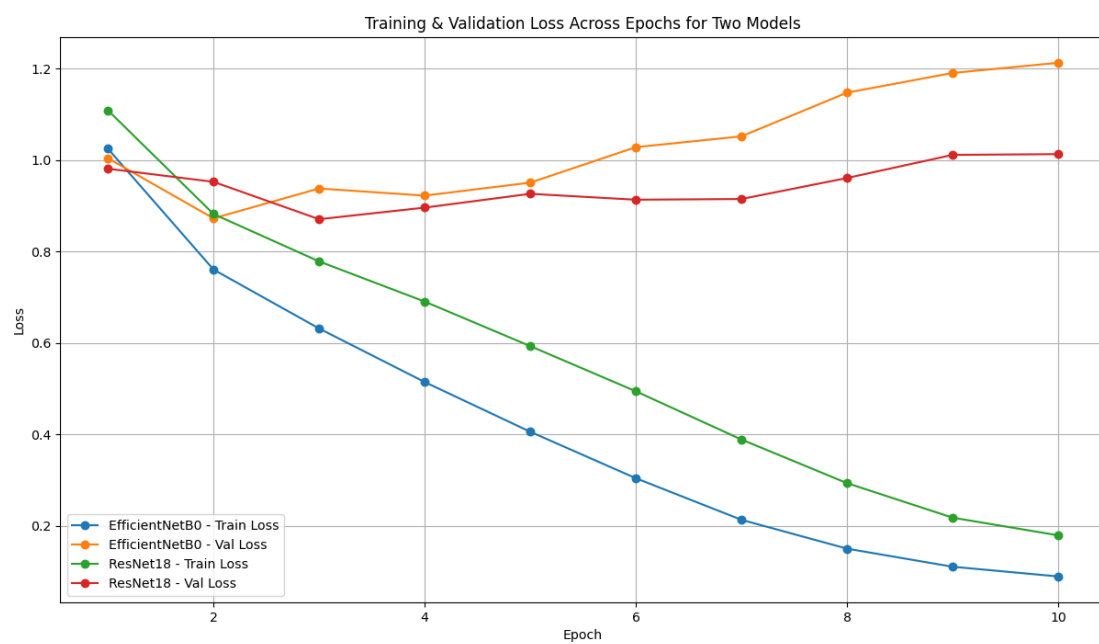


Figure 5.2: Notavbly, ResNet18 loss is lower.

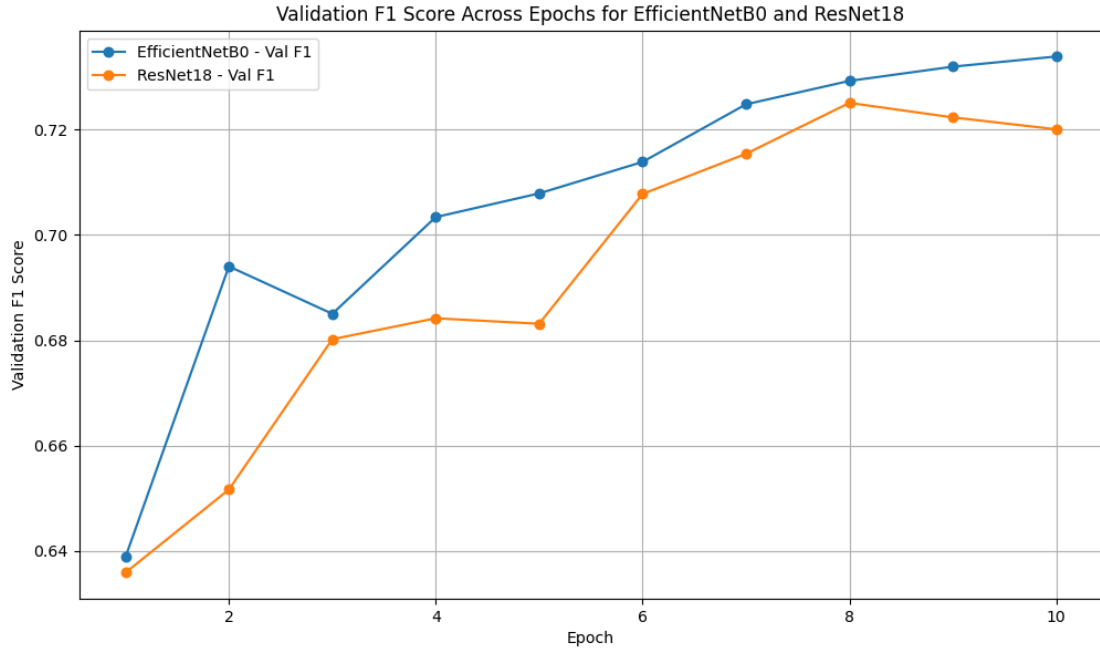


Figure 5.3: EfficientNetB0 takes the lead, as with validation accuracy.



Figure 5.4: Different representations of happiness in the dataset.

wanted to make our tests as close to reality as possible, and used to computers to show the pictures and the other one to run both student and educator and to show the emotions, predicted. Figure 5.5 shows how application has been tested.

The model's resulting accuracy was not as impressive, as we have imagined. On average accuracy was around 1-10%. This can be attributed to the fact, that

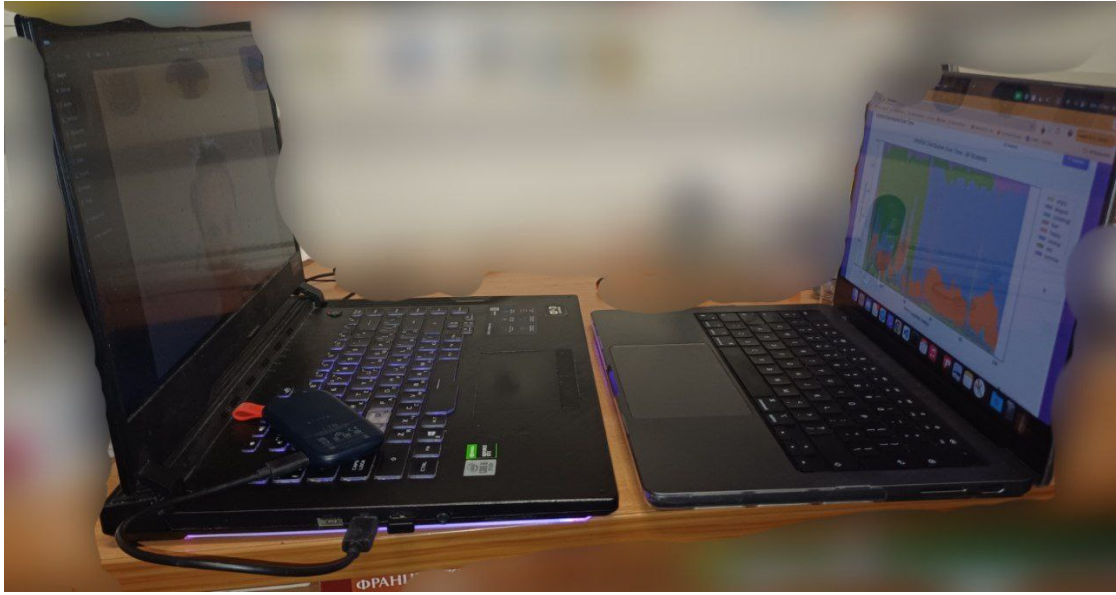


Figure 5.5: Set-up used to test application on MMO.

many of the pictures from MMO had their emotions shown with different expressions than those of the images in the training datasets. Figure 5.6 shows, for instance, the difference in disgust expressions between MMO and two other datasets. On the picture, Mr. Nosal does not show his teeth, as other two individuals. Instead, his lips are positioned tightly together. Nevertheless, there is some proximity with the expression of the gentleman on the left: they both have a wrinkle between the eyes.

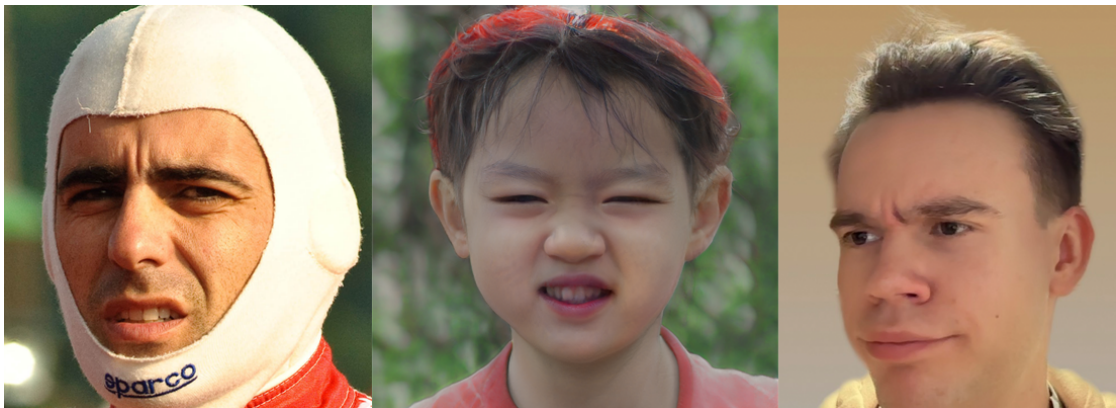


Figure 5.6: Different people show the same emotion slightly(or not) differently.

5.2 Recommendation of the Best Model

Although multiple models have shown solid performance, we have decided to use EfficientNetB0 in our application. This model has shown the best results during testing, even if by a small margin. The other models, such as ResNet50 and ResNet18 are also applicable, although less accurate. The VGG models are not really recommended, since they are really heavy and require much larger portion of time to train than their counterparts. The one thing that significantly stands out, while comparing this models, is the numbers of parameters.

While other models have tens of millions of parameters (VGG16 and VGG11 have more than 130 million parameters), EfficientNetB0 has around 5 million parameters, which makes it comparably compact. Table 5.1 shows the numbers of parameters across different models. The success of EfficientNetB0 in terms of accuracy is definitively linked to its size. As mentioned above, when models reach certain size, they begin to overfit, in other words, if the model is too large, it does not learn patterns in data, and instead simply "learns it by heart". Another important aspect of work with machine learning models, is how much resources in both time and computation they require. Since the models were trained using same computers, one can talk about resources to lesser extent in terms of computational resources, and more in terms of time needed, to reach overfitting, which signified that the model has reached its peak validation accuracy.

Validation accuracy is not, however, the sole measure of a model's performance. Such metrics as precision and Recall are also important while estimating the models. In this regard, the models performed similarly well, around 0.65 - 0.70 in both metrics, with the exception of EfficientNet-B0, which demonstrated both Recall and Precision of 0.82. Having considered the metrics, EfficientNet-B0 is a solid choice for an emotion recognition model.

Model	Parameters
VGG11	132,863,336 [17]
VGG16	138,357,544 [17]
ResNet-18	11,689,512 [18]
ResNet-50	25,557,032 [18]
EfficientNet-B0	5,300,000 [19]
EfficientNet-B3	12,200,000 [19]

Table 5.1: Numbers of parameters in different Convolutional Neural Networks.

5.3 Education Context

The ultimate goal of an educator is to provide the students with an enjoyable learning experience. It is, naturally, linked with emotions, experienced by the students [20]. If an Educator is not capable of understanding the Students' emotions, if they are unable to "read" the Students, this will lead to poorer quality of the education provided. If the students are feeling unpleasant emotions, such as anger, fear, disgust, it is not simply affecting the motivation, and the percentage of the students' absences, during classes [21], but also their performance [22]. If a student is not motivated at school, they do not really have a choice but to attend classes, even if the teachers are not investing enough effort. This leads to greater anxiety and depression among students. Young people rely on the teachers heavily at school, since they have not yet mastered the skill of self-learning. This is why, it is crucial that teachers, at schools in particular, make sure that the students are feeling positive emotions, such as curiosity, surprise and happiness, which all can be considered one emotion family, in our opinion. The other researchers [20] also propose such splits of emotions into families.

What is more, doing this, it is easier to determine the mood of the crowd, since such emotions as anger contempt and disgust are shown in similar way, but at the same time signal close emotions. It has been discussed to some extent the effects of poor emotional state on the younger students, it is also important to discuss them in the case of young adults. Students in universities, military academies, language courses, etc, have freedom to be absent whenever they do not wish to be present on the lectures/tutorials that cause distress. This leads to two outcomes. First of all, the students have worse performance, and secondly, they perpetuate this, by feeling apathetic to ask the Educator for improvements. Having considered related studies [22], the negative outcomes of negative emotions in education may be as follows: stress, low self-esteem, apathy towards learning, negative opinion of one's career choice.

The positive emotions may also have some effects that are not necessarily positive. For instance the researches [23] have shown that overly positive emotions can have an impact on the adequacy of the expectations of the students. It also has been noted that overly extatic feeling can negatively impact concentration and the understanding of the materials presented during the class.

5.4 Remarks and Conclusions

Having experimented with the application, we have made certain conclusion. Firstly, the application is depended on the network provider for providing stable working environment for it. We have encountered some issues with regard to the firewall

of our dermitory, since it blocked cloud service we used to connect Student and Educator during our tests. Another thing, that is wort mentioning is that it is important to consider multiple emotions togather, since many people show their emotions in a very similar manner. On the figure 5.7 one can see that. If two similar emotions are grouped together, for instance disgust and contempt, then the Educator could guess that the topic is unpleasant for the students. Similarly, we have seen that neutral emotion and anger are also very close, in general. The other important thing that affects accuracy of the application, is expressiveness of a particular person. If a person is naturaly calm, than it will be harder for a model to pick up signs of a particular emotion. The figure 5.8 shows different levels of expressiveness of happiness. To summarize, the application works properly if one considers emotion groups, and not single particular emotions.



Figure 5.7: Neutral emotion, fear, and surprise shown in a simmilar way.

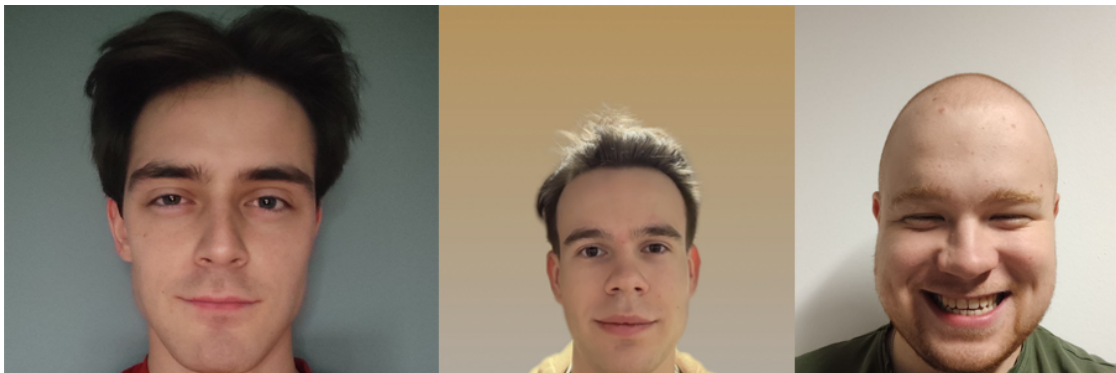


Figure 5.8: happiness shown with different levels of expressiveness.

Summary

In the project we have done, we have produced many interesting results. One of the most interesting ones are concerning model training. Not only we have used openly available datasets, we also managed to achieve reasonably high accuracies of emotion recognition. During the development of the application we have employed multitude of frameworks for backend and frontend. We had to reserve to such tools as Pinggy to establish communication between the Educator and Students.

The education context that we developed our application for, provided us with guidelines on what kind of users will be interacting with it. The resulting application uses web-based GUI, which was rated positively by the reviewers. The application is using a model, trained by us, which showed the highest validation accuracy amongst multiple models that we compared. The datasets used to train our model are of the reasonable size, consequently, other researchers can replicate our results without having to possess excessive storage. This is also important for the custom model-training, available in our application. The Users may choose different parameters, datasets, as well as choose which training metrics they would like to see during training. With the compact version of AffectNet, for instance, it is possible to train a model using far less space than with the full version.

During the development of our project, we have considered other solutions, available on the market. There are applications that help teachers/instructors to determine the general mood of school children through mobile UIs. They, however, rely on the students to indicate their emotions themselves. This may lead to students incorrectly signaling their emotions, due to various reasons, such as fear of not meeting the expectations. Our application relies on the model, so there is room for human-induced errors during emotion-recognition. Moreover, it does not save pictures of the Students, and it does not match Students to their emotions, preserving anonymity.

In the future it is possible to scale the application, including more datasets as well as larger and more precise models. This will lead to higher accuracy of emotion recognition. It is also possible to develop the idea of this project beyond educational context, as the demand grows on the systems, capable to understand humans better.

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